Akoizo Original Practice - Form 4

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Sample SD for 8,10,12 is…

A. 10

B. 8

C. 2

D. 4

Answer: _____

2. (2 points) A pilot primarily checks…

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Whether to delete awkward results

B. The final conclusion automatically

C. Every possible mechanism

D. Feasibility and procedure

Answer: _____

3. (2 points) Which addresses time-order drift?

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. Deleting late trials silently

B. Randomization or suitable blocking

C. Only more decimals

D. Only a larger title

Answer: _____

4. (2 points) A reproducible timing method needs…

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Only a final mean

B. Only a title

C. Explicit start and stop events

D. Only a stopwatch brand

Answer: _____

5. (2 points) A useful practice reflection leads to…

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. A specific correction tested on a new task

B. Only repeating the same answer

C. Blaming all variation on humans

D. Removing the error section

Answer: _____

6. (4 points) Write the example's claim and evidence.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

7. (4 points) Distinguish control condition and controlled variable.

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

8. (4 points) How can roles create contradictions?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

Answer: _____
_____
_____

9. (4 points) Why show with/without-outlier results?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

Answer: _____
_____
_____

10. (4 points) Why does replication not solve drift alone?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

Answer: _____
_____
_____

11. (2 points) Define “Range” in the context of Experimental Design.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: For response values 8, 10, 12, calculate mean, range, and sample SD.

Answer: _____
_____
_____

13. (2 points) Define “Independent variable” in the context of Experimental Design.

Answer: _____
_____
_____

14. (2 points) Define “Conclusion” in the context of Experimental Design.

Answer: _____
_____
_____

15. (2 points) Define “Sample standard deviation” in the context of Experimental Design.

Answer: _____
_____
_____

16. (2 points) Define “Replication” in the context of Experimental Design.

Answer: _____
_____
_____

17. (2 points) Define “Guard digits” in the context of Experimental Design.

Answer: _____
_____
_____

18. (2 points) Define “Extrapolation” in the context of Experimental Design.

Answer: _____
_____
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